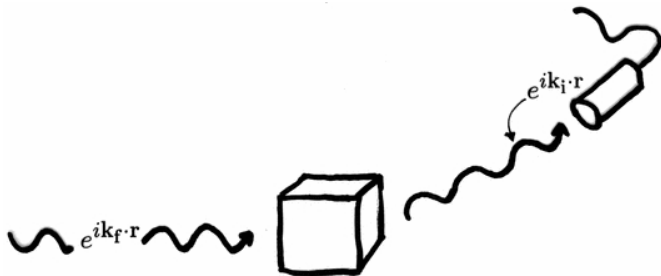


Introduction to QCr

Dylan Jayatilaka

MPI for Multidisciplinary Science, Goettingen; Chari group



Diffraction experiment on crystal with incoming and outgoing plane wave, and with detector.
(See the mistake?)

Elastic scattering:

Initial state: $e^{i\mathbf{k}_i \cdot \mathbf{r}} \psi_{\text{crystal}}$

Final state: $e^{i\mathbf{k}_f \cdot \mathbf{r}} \psi_{\text{crystal}}$

Introduction to QCr

1. What's in a name?
2. Quantum crystallography
3. The diffraction experiment & atomicity
4. Hohenberg-Kohn theorem
5. Hirshfeld atom refinement (`hart` demo)
6. X-ray wavefunction refinement
7. Some properties you can get

1. What's in a name?

Ewald's words when arguing for the IUCr

For any particular method, after having been applied for some time, tends either to become a routine method of observation, and as such of minor interest, or continually to *be merged with other methods supplementing it from fresh sides*.

1. What's in a name?

Ewald's words when arguing for the IUCr

For any particular method, after having been applied for some time, tends either to become a routine method of observation, and as such of minor interest, or continually to *be merged with other methods supplementing it from fresh sides*.

In the case of the X-ray method, this latter happened first when W. L. Bragg drew up a list of atomic radii. Later optical and magnetic observations, and the principles of atomic physics and of crystal chemistry which were being established, were added to the armoury supplementing the diffraction method.

1. What's in a name?

Ewald's words when arguing for the IUCr

For any particular method, after having been applied for some time, tends either to become a routine method of observation, and as such of minor interest, or continually to be merged with other methods supplementing it from fresh sides.

In the case of the X-ray method, this latter happened first when W. L. Bragg drew up a list of atomic radii. Later optical and magnetic observations, and the principles of atomic physics and of crystal chemistry which were being established, were added to the armoury supplementing the diffraction method.

Are not many . . . interested mainly in applying the results gained by diffraction methods **to the elucidation of chemical, mechanical and physical properties**? Is it not one of the main aims . . . **to link up X-ray diffraction with whatever chemical or physical methods seem promising to supplement it?**

Ewald, Nature (1944)

2. Quantum crystallography

2. Quantum crystallography

Quantum Crystallography seeks to “link up” X-ray diffraction with quantum mechanical calculation methods or other quantum mechanical models in order to:

- ▶ *Learn* something about the wavefunction or wavefunction-related entities composing the crystal,
- ▶ *Modify and improve* those wavefunctions, or entities;
- ▶ *Improved properties* obtained from the experiment.

(The last two are somewhat similar; since one only knows if the wavefunction is improved through the properties you can calculate from it).

$$\Psi \xrightarrow{\text{Fit to Expt1}} \Psi' \xrightarrow{\text{Predict Expt 2}} \text{Some property}$$

The primary way diffraction experiments link up with quantum mechanics and the wavefunction, is through various *densities*, especially the electron density ρ , which is closely associated to the wavefunction Ψ .

3. The diffraction experiment



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector. (See the mistake?)

3. The diffraction experiment



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector. (See the mistake?)

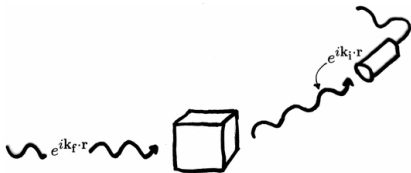
Elastic scattering:

Initial state: $e^{ik_i \cdot r} \psi_{\text{crystal}}$

Final state: $e^{ik_f \cdot r} \psi_{\text{crystal}}$

The experiment is just high-frequency Rayleigh scattering.

3. The diffraction experiment



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector. (See the mistake?)

Elastic scattering:

Initial state: $e^{ik_i \cdot r} \Psi_{\text{crystal}}$

Final state: $e^{ik_f \cdot r} \Psi_{\text{crystal}}$

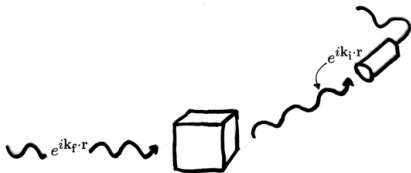
The experiment is just high-frequency Rayleigh scattering.

The treatment for scattering of other particles is conceptually the same, except you may have to consider the spin polarisation of the initial (and more rarely, the final) wave.

Different particles probe different properties e.g.

- photons, the electron density & local $\mathbf{B}$ field;
- neutrons, the nuclear density & local $\mathbf{B}$ field;
- electrons, the electrostatic potential & local $\mathbf{B}$ field;

3. The diffraction experiment



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector. (See the mistake?)

Elastic scattering:

Initial state: $e^{i\mathbf{k}_i \cdot \mathbf{r}} \Psi_{\text{crystal}}$

Final state: $e^{i\mathbf{k}_f \cdot \mathbf{r}} \Psi_{\text{crystal}}$

The experiment is just high-frequency Rayleigh scattering.

The treatment for scattering of other particles is conceptually the same, except you may have to consider the spin polarisation of the initial (and more rarely, the final) wave.

Different particles probe different properties e.g.

- photons, the electron density & local $\mathbf{B}$ field;
- neutrons, the nuclear density & local $\mathbf{B}$ field;
- electrons, the electrostatic potential & local $\mathbf{B}$ field;

Probability to see an X photon (Born rule):

$$\begin{aligned} & \left| \langle e^{i\mathbf{k}_i \cdot \mathbf{r}} \Psi_{\text{crystal}} | e^{i\mathbf{k}_f \cdot \mathbf{r}} \Psi_{\text{crystal}} \rangle \right|^2 \\ &= \left| \int e^{i(\mathbf{k}_f - \mathbf{k}_i) \cdot \mathbf{r}} |\Psi_{\text{crystal}}|^2 \right|^2 \\ &\approx \left| \int e^{i\mathbf{k} \cdot \mathbf{r}} \rho(\mathbf{r}) \right|^2 \\ &\equiv |F(\mathbf{k})|^2 \end{aligned}$$

structure
factor

and where

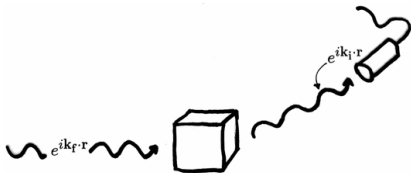
$$\mathbf{k} = \mathbf{k}_f - \mathbf{k}_i$$

scattering
vector

$$\rho(\mathbf{r}) \approx |\Psi_{\text{crystal}}|^2(\mathbf{r})$$

electron
density

3. The diffraction experiment



X-ray experiment on crystal with incoming and outgoing plane wave, and with detector. (See the mistake?)

Elastic scattering:

Initial state: $e^{ik_i \cdot r} \Psi_{\text{crystal}}$

Final state: $e^{ik_f \cdot r} \Psi_{\text{crystal}}$

The experiment is just high-frequency Rayleigh scattering.

The treatment for scattering of other particles is conceptually the same, except you may have to consider the spin polarisation of the initial (and more rarely, the final) wave.

Different particles probe different properties e.g.

- photons, the electron density & local $\mathbf{B}$ field;
- neutrons, the nuclear density & local $\mathbf{B}$ field;
- electrons, the electrostatic potential & local $\mathbf{B}$ field;

Probability to see an X photon (Born rule):

$$\begin{aligned} & \left| \langle e^{ik_f \cdot r} \Psi_{\text{crystal}} | e^{ik_i \cdot r} \Psi_{\text{crystal}} \rangle \right|^2 \\ &= \left| \int e^{i(\mathbf{k}_f - \mathbf{k}_i) \cdot \mathbf{r}} |\Psi_{\text{crystal}}|^2 \right|^2 \\ &\approx \left| \int e^{i\mathbf{k} \cdot \mathbf{r}} \rho(\mathbf{r}) \right|^2 \\ &\equiv |F(\mathbf{k})|^2 \end{aligned}$$

structure
factor

and where

$$\mathbf{k} = \mathbf{k}_f - \mathbf{k}_i$$

scattering
vector

$$\rho(\mathbf{r}) \approx |\Psi_{\text{crystal}}|^2(\mathbf{r})$$

electron
density

Since $F(\mathbf{k})$ is a Fourier transform, and since the crystal $\rho(\mathbf{r})$ is periodic only certain scattering directions are allowed: those where scattering vector $\mathbf{k}$ forms a “reciprocal lattice” and which lies on the Ewald sphere.

These are seen as diffraction spots, and their direction and intensity $I \propto |F(\mathbf{k})|^2$ is measured.

3. The diffraction experiment

Atomicity: a key assumption

Many (all?) of the models in this workshop decompose the probed density as a sum of atomic densities.

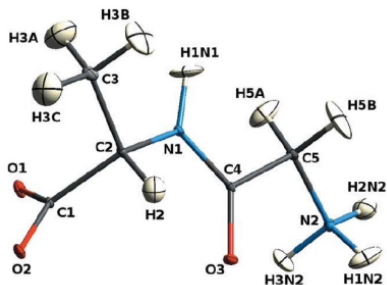
$$\rho(\mathbf{r}) = \sum_A^{n_{\text{atoms}}} \rho_A(\mathbf{r})$$

There is no fundamental reason to do this.

– But it simplifies the way nuclear vibrations are treated –
atomic densities can be “smeared” by the atomic
displacement parameters (ADPs).

3. The diffraction experiment

Atomic displacement parameters



The atoms in crystal structures appear like **Australian footballs** — because their position is imprecise.

Figure 1: Structure of Gly-L-Ala determined by X-ray diffraction at 12K showing the probability distributions associated with the nuclear positions. Even in a perfect crystal at low temperature these are present and indicate that, like the electrons, the nuclei should also be described by (phonon) wavefunctions or density matrices. From Capelli et al (2014).

3. The diffraction experiment

Atomic displacement parameters

The atoms in crystal structures appear like **Australian footballs** — because their position is imprecise.

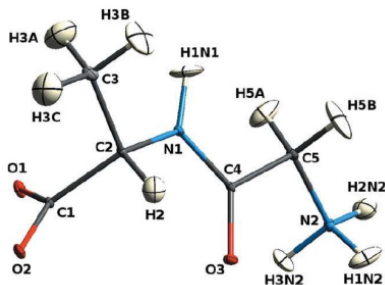


Figure 1: Structure of Gly-L-Ala determined by X-ray diffraction at 12K showing the probability distributions associated with the nuclear positions. Even in a perfect crystal at low temperature these are present and indicate that, like the electrons, the nuclei should also be described by (phonon) wavefunctions or density matrices. From Capelli et al (2014).



3. The diffraction experiment

Atomic displacement parameters

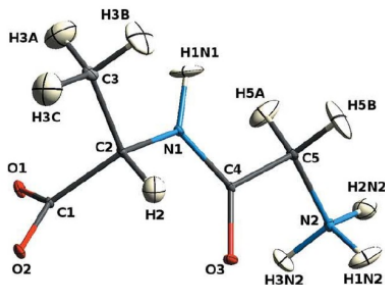


Figure 1: Structure of Gly-L-Ala determined by X-ray diffraction at 12K showing the probability distributions associated with the nuclear positions. Even in a perfect crystal at low temperature these are present and indicate that, like the electrons, the nuclei should also be described by (phonon) wavefunctions or density matrices. From Capelli et al (2014).

The atoms in crystal structures appear like **Australian footballs** — because their position is imprecise.

The reason is twofold:

— that *the nuclei are wavelike particles* and are delocalised, we should model their probability distributions.

— the nuclei in different unit cells don't have the same position, they are not quite translationally equivalent, *the crystallites are misaligned*.

4. The Hohenberg-Kohn theorem

Why the obsession with the electron density?

4. The Hohenberg-Kohn theorem

Why the obsession with the electron density?

- ▶ Hohenberg & Kohn (1964) proved (in four lines!) that the QM energy is a unique functional of the electron density.
- ▶ Later, Kohn & Sham (1965) defined a single determinant wavefunction Ψ_{KS} which had the “exact” Born-Oppenheimer electron density.
- ▶ Using Ψ_{KS} they approximated the kinetic energy very well, which allowed to develop good approximate energy functionals.

This is called “density functional theory” (DFT) a development which led to the Nobel prize in 1999.

We shall now show that conversely $v(\mathbf{r})$ is a unique functional of $n(\mathbf{r})$, apart from a trivial additive constant.

The proof proceeds by *reductio ad absurdum*. Assume that another potential $v'(\mathbf{r})$, with ground state Ψ' gives rise to the *same* density $n(\mathbf{r})$. Now clearly [unless $v'(\mathbf{r}) - v(\mathbf{r}) = \text{const}$] Ψ' cannot be equal to Ψ since they satisfy different Schrödinger equations. Hence, if we denote the Hamiltonian and ground-state energies associated with Ψ and Ψ' by H, H' and E, E' , we have by the minimal property of the ground state,

$$E' = \langle \Psi', H' \Psi' \rangle < \langle \Psi, H' \Psi \rangle = \langle \Psi, (H + V' - V) \Psi \rangle,$$

so that

$$E' < E + \int [v'(\mathbf{r}) - v(\mathbf{r})] n(\mathbf{r}) d\mathbf{r}. \quad (6)$$

Interchanging primed and unprimed quantities, we find in exactly the same way that

$$E < E' + \int [v(\mathbf{r}) - v'(\mathbf{r})] n(\mathbf{r}) d\mathbf{r}. \quad (7)$$

Addition of (6) and (7) leads to the inconsistency

$$E + E' < E + E'. \quad (8)$$

Thus $v(\mathbf{r})$ is (to within a constant) a unique functional of $n(\mathbf{r})$; since, in turn, $v(\mathbf{r})$ fixes H we see that the full many-particle ground state is a unique functional of $n(\mathbf{r})$.

The Hohenberg & Kohn (1964). It does not give any idea *how* to obtain the magical functional.

Repository and workshop

`github.com/dylan-jayatilaka/tonto/`

Look in the `docs/` folder.

5. Hirshfeld atom refinement

In Hirshfeld Atom Refinement the Goodness of Fit (squared)

$$\text{GoF}^2 = \frac{1}{N_{\text{refl}} - N_{\text{param}}} \sum_{r=1}^{N_{\text{refl}}} \left[\frac{s|F_r^{\text{calc}}| - F_r^{\text{obs}}}{\sigma_r^{\text{obs}}} \right]^2$$

is minimized with respect to position parameters and ADPs, and a scale factor s .

F_k^{calc} depends on an approximate wavefunction Φ , as follows.

5. Hirshfeld atom refinement

In Hirshfeld Atom Refinement the Goodness of Fit (squared)

$$\text{GoF}^2 = \frac{1}{N_{\text{refl}} - N_{\text{param}}} \sum_{r=1}^{N_{\text{refl}}} \left[\frac{s |F_r^{\text{calc}}| - F_r^{\text{obs}}}{\sigma_r^{\text{obs}}} \right]^2$$

is minimized with respect to position parameters and ADPs, and a scale factor s .

F_k^{calc} depends on an approximate wavefunction Φ , as follows.

Aspherical atomic densities ρ_A are “cut out” of the total density and are Fourier transformed (numerically) to give aspherical atomic form factors f_A for each atom A .

The f_A and wavefunction are recalculated every time the structure is refined i.e. the structure factors depend on the structure.

So the minimization has to be done iteratively.

5. Hirshfeld atom refinement

Formula for the structure factor from a determinant wavefunction Φ

$$F_k^{\text{calc}}[\Phi] = s |F^{\text{model}}(\mathbf{k})|$$
$$F^{\text{model}}(\mathbf{q}_k) = \sum_A^{\text{atoms}} n_A^{-1} \sum_{\{\mathbf{S}, \mathbf{t}\}}^{\text{symops}} \bar{f}_A(\mathbf{S}^T \mathbf{k}) e^{i\mathbf{k} \cdot \mathbf{R}_A} e^{i\mathbf{k} \cdot \mathbf{t}}$$

where:

s	=	Scale factor
$\mathbf{k}$	=	k -th scattering vector = $2\pi \mathbf{A} \mathbf{h}$
$\{\mathbf{R}_A\}$	=	Asymmetric atom coordinate
$\{\mathbf{S}, \mathbf{t}\}$	=	Roto-translational symmetry operators
n_A	=	Atom site symmetry factor
$\bar{f}_A(\mathbf{q}_k)$	=	Thermally-averaged aspherical form factor
	=	$f_A(\mathbf{k}; \{\mathbf{R}_A\}; \Phi) \times \text{TF}(\mathbf{k})$
$f_A(\mathbf{k}; \{\mathbf{R}_A\}; \Phi)$	=	Static aspherical form factor; from Φ
$\text{TF}(\mathbf{k})$	=	Temperature factor (Gram-Charlier form)
	=	$e^{-\frac{1}{2!} \mathbf{U}^2 : \mathbf{k}} \left[1 - \frac{i}{3!} \mathbf{U}^3 : \mathbf{k} + \frac{1}{4!} \mathbf{U}^4 :: \mathbf{k} \right]$
$\mathbf{U}^2, \mathbf{U}^3, \mathbf{U}^4$	=	Atomic displacement parameters, ADPs

6. X-ray wavefunction refinement (XWR)

How we fit Φ with not enough data

6. X-ray wavefunction refinement (XWR)

How we fit Φ with not enough data

- ▶ We do a *constrained fit*.

In crystallography, this is also called a *restrained fit*.

- ▶ We minimize the Lagrangian

$$L = E_{QM}[\mathbf{c}] + \lambda \text{GoF}^2[\mathbf{c}]$$

with respect to the orbital coefficients $\mathbf{c}$.

- ▶ The term λGoF^2 is the least-squares penalty; it enforces the fitting constraint.

6. X-ray wavefunction refinement (XWR)

How we fit Φ with not enough data

- ▶ We do a *constrained fit*.

In crystallography, this is also called a *restrained fit*.

- ▶ We minimize the Lagrangian

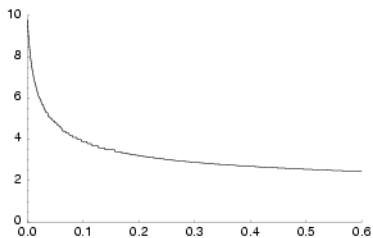
$$L = E_{QM}[\mathbf{c}] + \lambda \text{GoF}^2[\mathbf{c}]$$

with respect to the orbital coefficients $\mathbf{c}$.

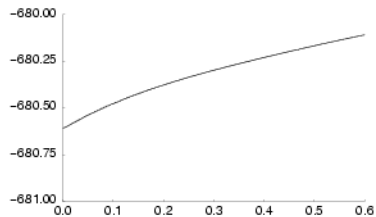
- ▶ The term λGoF^2 is the least-squares penalty; it enforces the fitting constraint.

Note:

- ▶ Choosing $\lambda = 0$ gives a normal QM calculation.
- ▶ Choosing $\lambda = \infty$ gives a “regularized” least squares fit.
- ▶ In principle we can get a perfect fit!
- ▶ In principle we need no experimental data.
(But then, we do not have an experimental model!).



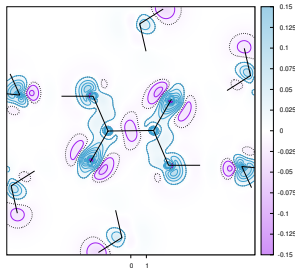
GoF^2 vs λ



E vs λ

7. Some properties you can get

From Davidson, Grabowsky & Jayatilaka (2022)

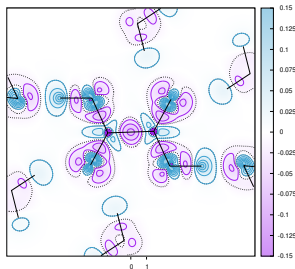


Effect on ED do to fitting HF wavefunction simultaneously to 12 data different sets.

Contours at $0.025 e \text{ \AA}^{-3}$.

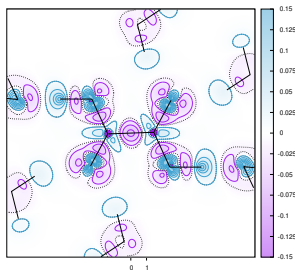
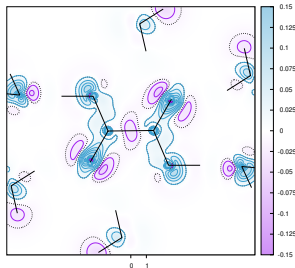
Top: XCHF wavefunction

Bottom: Calculation: CCSD-HF with pt. charges



7. Some properties you can get

From Davidson, Grabowsky & Jayatilaka (2022)



Effect on ED do to fitting HF wavefunction simultaneously to 12 data different sets.

Contours at $0.025 \text{ e } \text{\AA}^{-3}$.

Top: XCHF wavefunction

Bottom: Calculation: CCSD-HF with pt. charges

Good qualitative agreement of electron density changes ' due to crystal and correlated effects.

—Except near the atom centers.

— CC bond e- depletion is exaggerated in theory.

— H atom e- buildup persists in theory but is now much smaller.

Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)

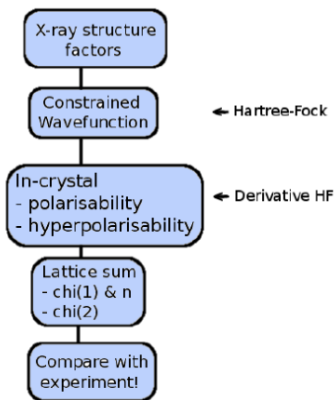
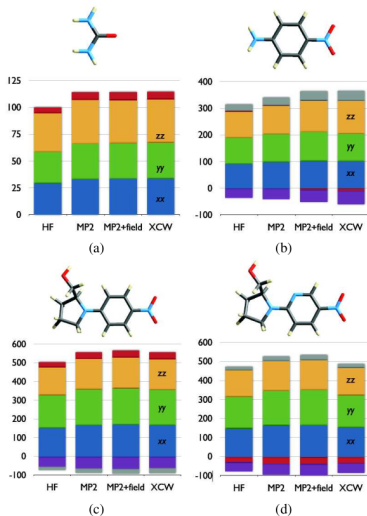


Fig. 11 Outline scheme for calculating refractive indices and NLO properties from X-ray data. See ref [15] for details.

Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)



Molecular polarizabilities and refractive indices

Jayatilaka, Munshi, Turner, Howard, Spackman (2009)

		MP2				
	axis	HF	MP2	+field	XCW	Expt
urea	$n_a = n_b$	1.39	1.44	1.45	1.45	1.477
	n_c	1.52	1.62	1.59	1.60	1.583
POM	n_a	1.55	1.58	1.58	1.55	1.625
	n_c	1.58	1.62	1.62	1.60	1.663
	n_b	1.71	1.82	1.81	1.78	1.829
NPP	n_X	1.42	1.45	1.44	1.45	1.457
	$n_Y = n_b$	1.63	1.69	1.70	1.67	1.774
	n_Z	1.71	1.80	1.82	1.79	1.829
PNP	n_X	1.43	1.45	1.45	1.43	1.456
	$n_Y = n_b$	1.62	1.68	1.69	1.61	1.732
	n_Z	1.70	1.80	1.82	1.74	1.880

Refractive indices n .

Don't look: the results from fitted wavefunctions are mostly better when compared to experiment or better calculations.

Other QCr talks & software in this workshop

- ▶ “Multipoles” to model aspherical atomic densities seen in proteins.
 - The multipoles have a radial part derived from atomic wavefunctions.
 - TAAM (Dominiak) - multipole parameters fitted to quantum chemical calculations (Dominiak)
 - **MoPro** - multipole parameters fitted to experiment (Jelsch, Guillot)
- ▶ Hirshfeld atom refinement (HAR)
 - Aspherical atomic densities partitioned directly from quantum chemical ρ
 - **Tonto** & **IamaGOET** - chemical crystallography (Me)
 - **Olex2** - chemical crystallography (Kleemiss)
 - **DiSCaMB** - Also targeting molecular biology. (Chodkiewicz)
- ▶ Structural “dynamic” (T dependent?) QCr
 - Atomic displacement parameters (ADPs) and thermodynamics! (Hoser & Madsen)
- ▶ Neutrons!
 - Single crystals (Wang)
 - Polarised neutron diffraction (Cao)
 - Materials (Kaneko)
 - Dynamic nuclear (spin) polarisation for H atoms (Myles)
 - Combined X-ray & neutron analysis (Cao)
 - **Cryspy** - Spin densities! (Marshall)